

Checkpoint Simulation

Unity 3D Simulation & Operational Optimization Analysis

Problem Overview

The Challenge



Increased Holiday Volume: Holiday travel surge increases checkpoint volume, causing long queues and wait times.



Operational Evaluation: Need to evaluate operational changes before peak season.

Strategic Objective

Build a discrete-event simulation to test staffing/lane configurations, predict performance, and recommend cost-effective improvements

Key Question

Where is the bottleneck – security or immigration? How many additional counters are needed?

| Simulation Design & Assumptions

Core Mechanics

Flow: Arrival → security queue → security processing → immigration queue → immigration (manual/automated) → exit.

Arrival Model: Poisson process, rate extracted from video analysis (1.3 persons/s, 95% main flow).

Flow Rules: Sequential logic ensuring security is cleared prior to immigration.

Assumptions

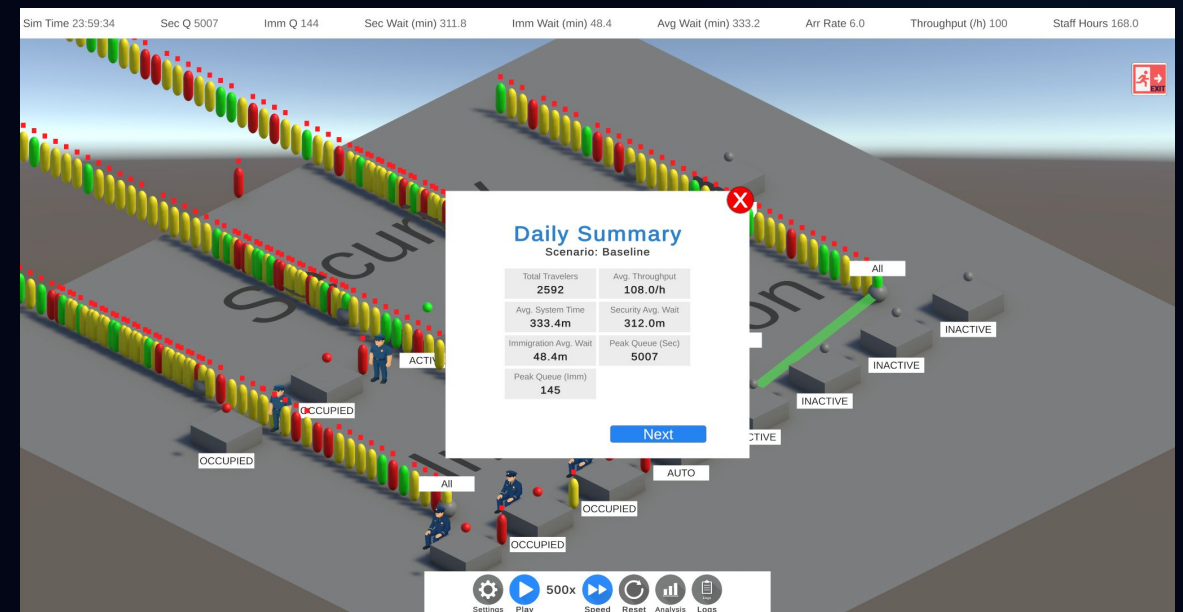
- FIFO queues within lanes.
- Triangular processing times (min, mode, max) with optional enhanced screening.
- Automated e-gates serve only citizens; 5% error rate.
- Time-of-day multipliers (8 blocks).
- Traveler types affect walking speed & processing time (standard at base speed and processing time; business +20% speed/-30% time; family opposite).

Baseline System Performance

Configuration: 3 security counters, 4 immigration counters (all manual), arrival rate: 4, security processing (base/min/max: 60/30/90; enhanced screening (takes longer): 5%), immigration processing (base/min/max: 90/45/135; enhanced screening: 10%).

Bottleneck: Immigration counters at 81% utilization, security only 77%. Queue builds at immigration.

Improvements: Adjust the processing time at respective counters to improve clearance rate



*End-of-day logging latency identified; optimization can be done for future build

Scenario Comparison

Scenarios	Sec.,Imm. Counters	Avg. Processing Time	Throughput/hr
Baseline	3, 4	60, 90	108 travelers/hr
Holiday Rush	6, 8	60, 90	179 travelers/hr
Optimized	5, 6	50, 75	167 travelers/hr
Wet Weather	3, 4	75, 105	95 travelers/hr

Simulation Stress Note

'Optimized' and 'Holiday Rush' scenarios involve heavy traffic volumes. To handle the huge crowd, it is best to maxed-out counters. However, these runs are significantly performance-intensive due to the high density of active traveler agents.

Recommendations & Operational Insights

Tech Roadmap

Implement **Unity DOTS and ECS** in future builds to significantly improve performance for high-density scenarios.

Traveler Logs can use virtual scrolling to improve performance when opening panel.

Resource Allocation

Prioritize **Security**. Adding 2 counters as long queues backed up at Security in all scenarios. In real-life scenario, there are often limited security counters, and not all travelers are screened.

Key Takeaway

Simulation enables **data-driven staffing**. We can verify bottlenecks to optimise checkpoint traffic. Security does not screen everyone at peak, otherwise, more sec. counters are needed

Sensitivity Note: Results are sensitive to processing time distributions. Refinement with real-time data is recommended for greater predictive accuracy.